2×2 Mesh NoC: viva quick reference

Project: four-router, 32-bit single-flit XY NoC in VHDL-2008. **Target:** xc7s15cpga196-2, Vivado 2026.1/XSim; local target runs pending.

Question	Precise answer
Why a NoC?	Packet links separate communication from endpoint functions; this project is a small implementation study, not proof of universal superiority over a bus.
Which coordinates?	R00(0,0), R01(1,0), R10(0,1), R11(1,1); X east, Y south; index=X+2Y.
Which route?	X first, then Y, then Local. 0→3 is 0→1→3; 2→1 is 2→3→1.
When does a packet move?	Rising edge, reset inactive, external valid AND ready. Data/valid hold during a stall.
Why two valid fields?	Bit 31 is a legal-packet marker; the external valid signal governs handshake.
Which buffering?	Three depth-four input FIFOs per corner, twelve total; no bypass or virtual channels.
How is output stable?	Held one-hot grant retains the selected FIFO head; pop occurs only on acceptance.
What happens at a full boundary?	A simultaneous pop is accepted but push is refused; the producer retries next cycle.
What limits throughput?	One arbitration bubble after each acceptance gives 0.5 flits/cycle per output; independent outputs can act together.
What fairness is guaranteed?	A persistent requester is served within five successful grants in the five-port arbiter test, assuming progress; stalled cycles do not count.
Is reset packet loss?	It deliberately flushes storage. Depth-four directed accounting is 106=74 delivered+32 reset discards; no unexplained pending packets.
What was actually tested?	Fourteen GHDL test configurations and 93 RTL experiments; 82,523 accepted benchmark packets all delivered.
What is latency?	Delivery-accept edge minus injection-accept edge. Empty/ready local=2, one-hop=4, two-hop=6 cycles.
Do deeper FIFOs improve saturation?	More accepted backlog raises latency; measured hotspot output remains 0.5 flits/cycle for depths 2/4/8.
Is 10 ns a measured clock?	No. It is an XDC design target; no fMAX, board oscillator or bits/s is established.
Are CPU/AI blocks real?	No; synthesizable finite traffic models only. No Zynq PS/AXI/Linux subsystem.
What remains?	XSim, exact-device utilization/STA, verified board constraints/bitstream and secure private GitHub push.

Evidence boundary: GHDL checks are supplemental. A green dashboard/CI does not certify FPGA timing. VHDL timing simulation is unsupported in Vivado 2026.1; use functional simulation and routed STA. No FPGA power or bus-improvement claim was measured.